



KKBOX AI Playlist Usage Insights and New Positioning

December, 24, 2025

KKBOX

AGENDA

01 | Abstract

02 | Current Status

03 | User Segmentation

04 | Focused Analysis

05 | Strategy Design

06 | Summary

Executive Abstract

Goal	Improve Collection Rate (CR) and Return Rate (RR) , which are used to measure satisfaction.		
Expected result	Improve user satisfaction	Explore new application scenarios	
Current	There is a gap in understanding between the AI and the user's prompts, causing users to feel that the playlist content does not meet their expectations. Users lack the motivation to save and reuse AI playlists, leading to low save and return rates.		
Analyze Direction	Based on system context and user behavior, explores the use of Prompt tags and semantic analysis.		
Pain Point	Users aren't good at inputting structured commands and natural language descriptions	Users use AI playlist as general search function	
Solution	Guided Prompt	AI music consultant	AI community

02 Current Status

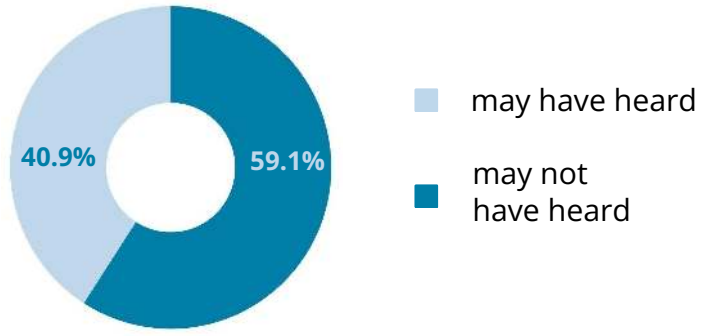


Al-generated playlist does not meet user expectationsThe key is to balance personalized recommendations with new song discovery.

User Journey	Current	Data support
During	AI-Prompt: Understanding the Gap	a gap of 1.24 for songs matching needs a gap of 1.07 for songs matching musical styles
After	The playlist did not meet expectations	Average collection rate: 20.50% The reflux rate was only 17.1%.

59.1% of users want more unfamiliar songs recommended

Q. Would you prefer playlist includes songs you haven't heard before or songs you might have heard?



User needs It must be based on my profile (personalized), and also provide songs that I wouldn't normally actively search for (surprise discovery)

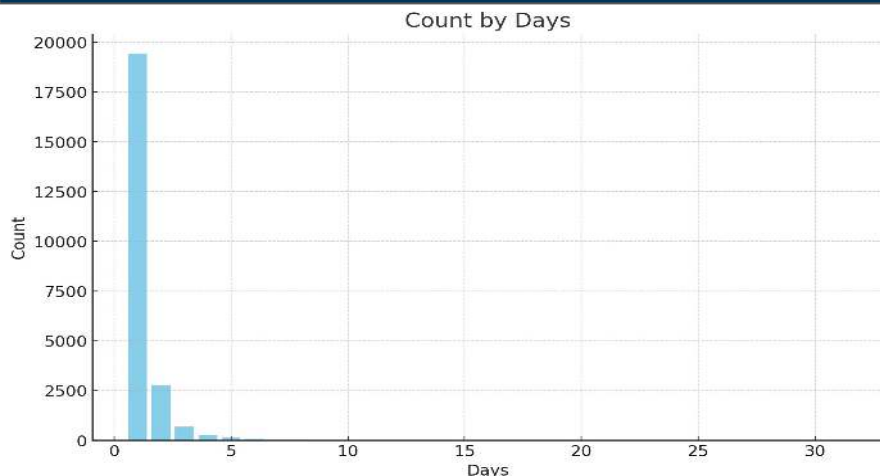
03 User Segmentation



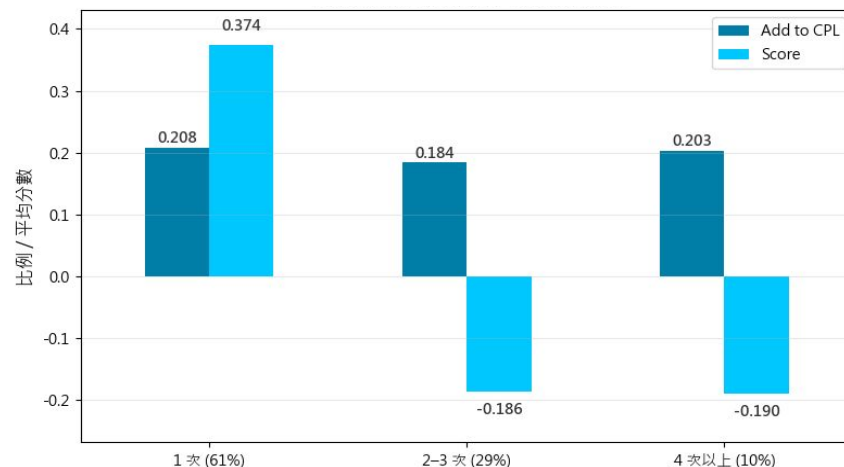
High interaction failed to generate satisfaction, indicating highly differentiated user needs

- Users lack motivation to return: 82.9% of users used it only once, 11.8% used it twice, and only 5.3% used it more than twice. The average satisfaction score for AI Search was negative, indicating that AI playlists are unlikely to establish a sustained willingness to use them.
- Deeper interaction diminishes the experience: Negative reviews accounted for 54.8%; and the collection rate decreased with increased interaction.

AI playlist usage days



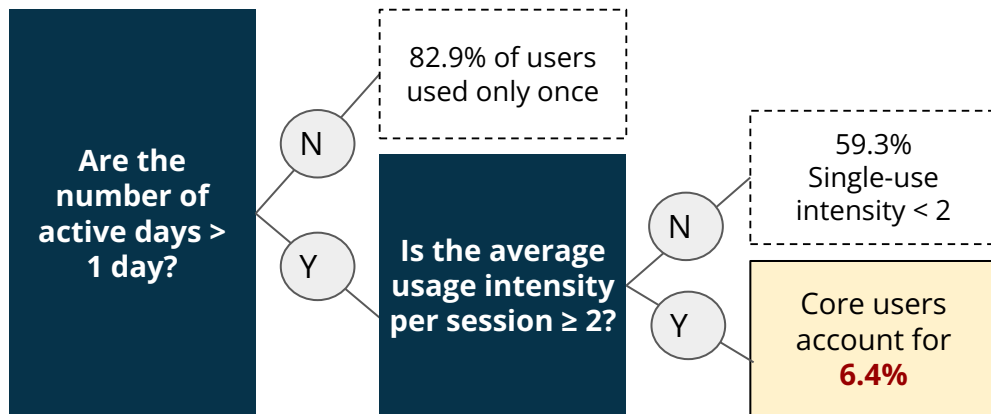
Consecutive usage count and adding to playlist



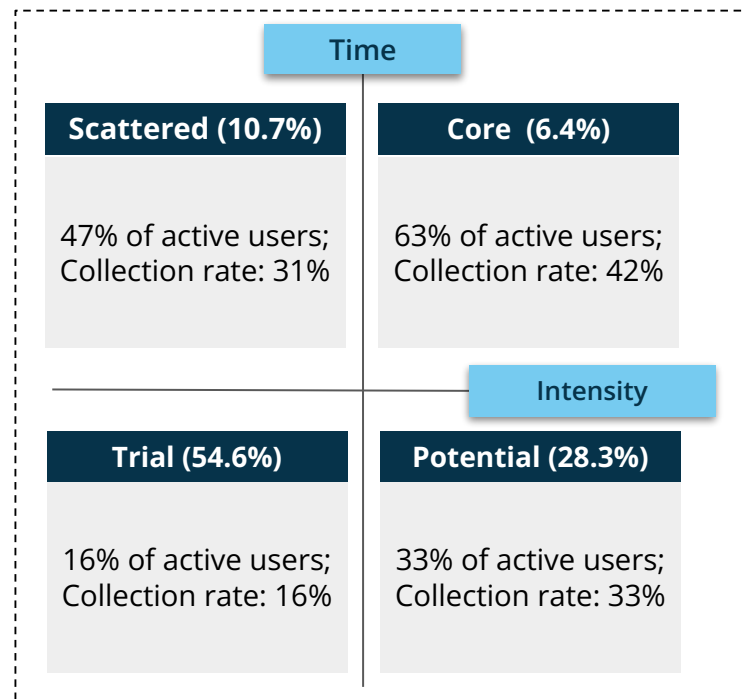
Users were segmented by Time x Interaction intensity to explore behavioral characteristics and differences.

Spindle

Based on their level of engagement, Established using "time dimension" and "intensity of single interaction"

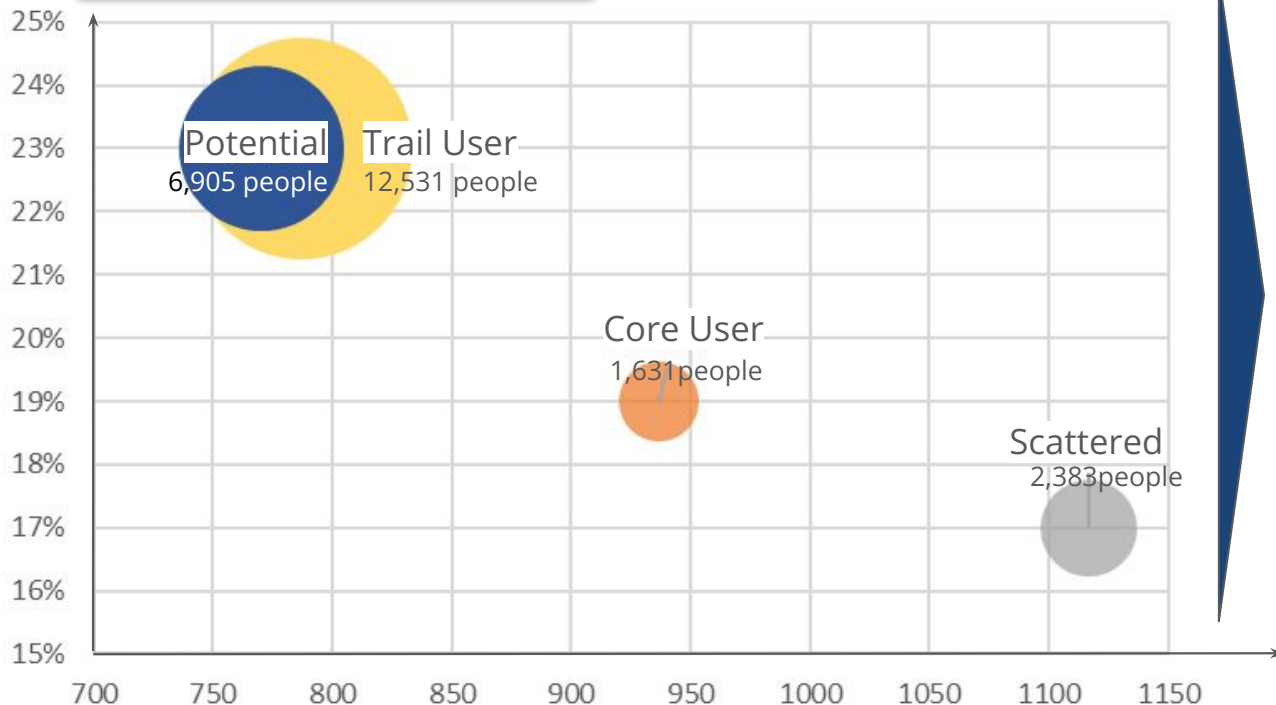


Index	Definition	Type
Distinct days	Number of active days	• 2 days or more • 1 day
Intensity	prompt_count / session_count	• ≥ 2 • Less than 2



Core and Scattered users enjoy exploration; Potential and Trail users prefer personalized recommendations.

Depth (Top 10 Concentration)



Potential and Trail users

Large user base, high concentration, and low breadth of preferences

>> Preferred Personalized

Core and Scattered users

Few users, low concentration, high breadth of listening preferences

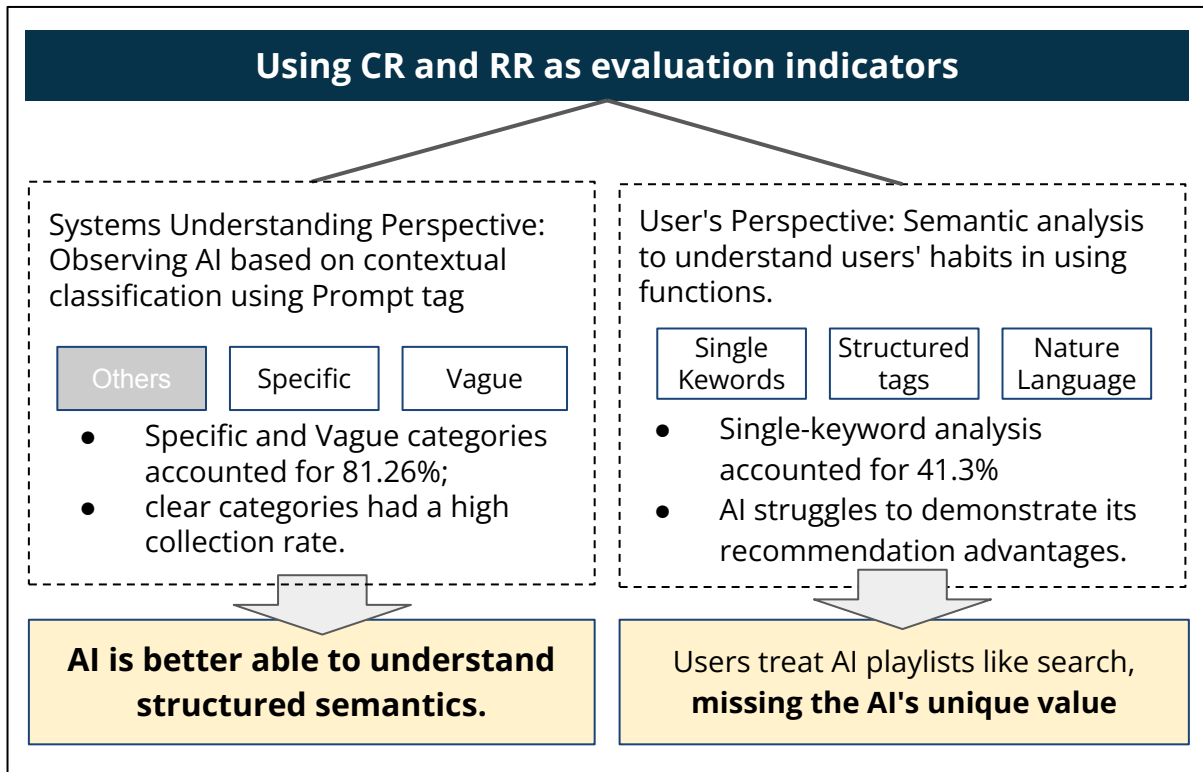
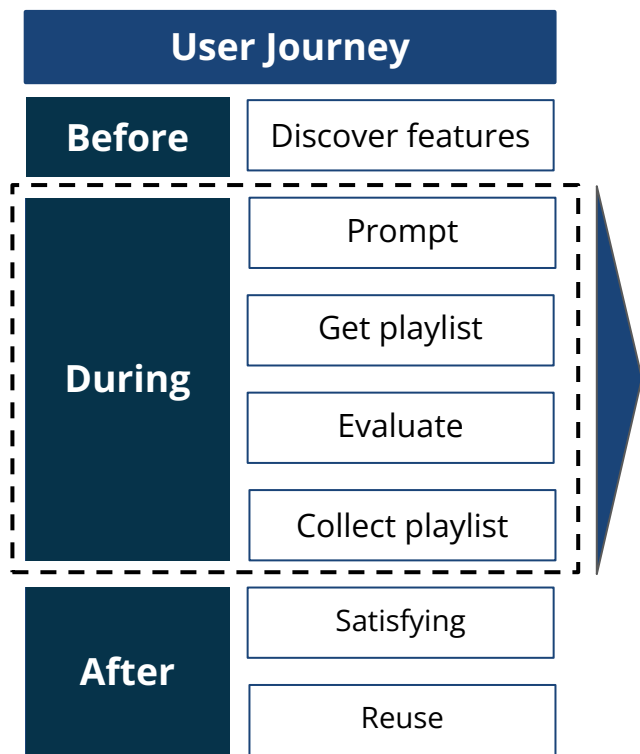
>> Preferred Exploration

Breadth (song diversity)

03 Focused Analysis



Analyzing Prompt behavior from both the system and user perspectives addresses the core pain point

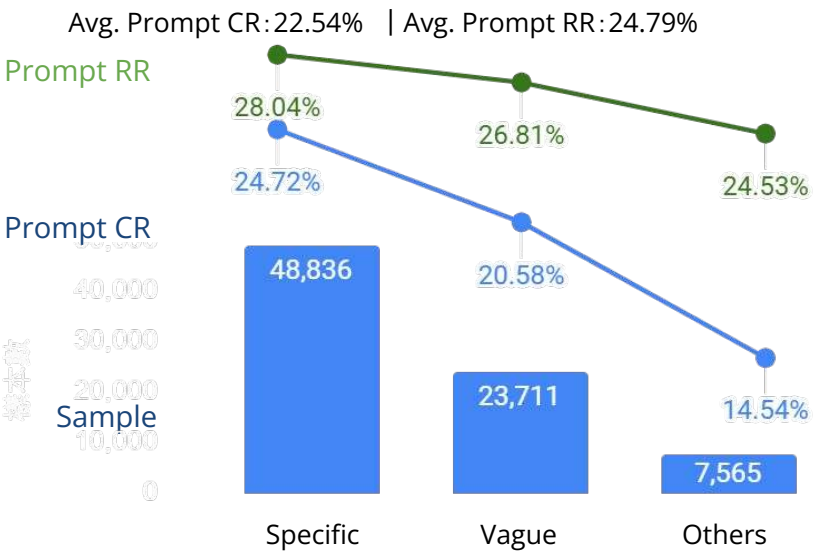


Vague prompts hinder AI comprehension; structured semantics accurately map music features, boosting retention and return rates

Prompt Classification: By Search Context

cat.	Directions	Ex.
Specific	- System-defined tags enable direct AI retrieval - High CR & RR indicate that AI excels at processing it	Artist, Musical style, Region
Vague	Absence of system-defined tags prevents direct search retrieval	Mood, Situation, Movie, Drama
Neutral	with a small sample size and low levels of CR & RR	others

Summary of analysis results



Problem summary

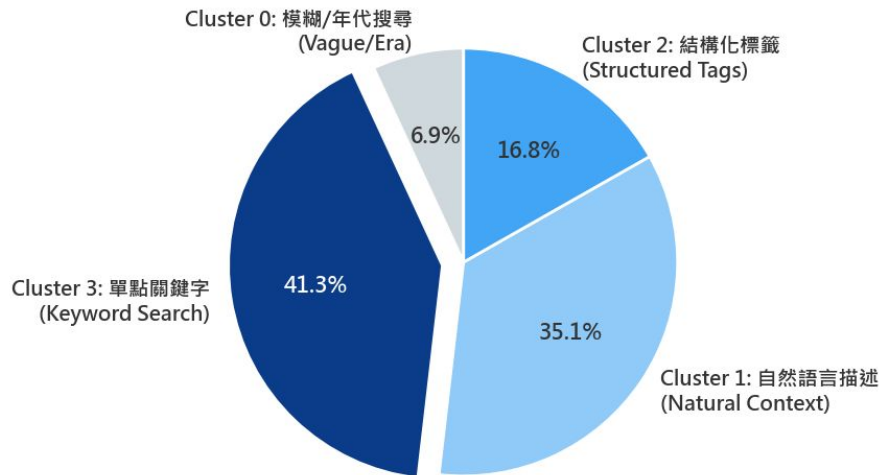
- Structured prompts drive higher retention by aligning better with music features
- Structured semantics map directly to music features, driving higher RR and CR

SBERT and K-Means clustering reveal that users predominantly input single-keyword prompts

Analysis process

- Data Cleaning: Filtered out system suggestions, leaving 45,583 organic user prompts.
- Vectorization (SBERT): Converted prompts into semantic vectors to capture contextual info.
- K-Means Clustering: Categorized inputs into four types
- Analysis: Named clusters and analyzed save rates vs. word counts to derive actionable insights.

User percentage by behavioral pattern



Scale and usage of each behavior

Cluster 0: Vague

「90s」、「以前的歌」、「懷舊」

N = 3,143

Cluster 1: Natural Context

「想聽輕鬆英文歌來讀書」、「運動時節奏不要太快」

N = 15,980

Cluster 2: Structured Tags

「輕快 | 英文 | 女生 vocal」

N = 7,648

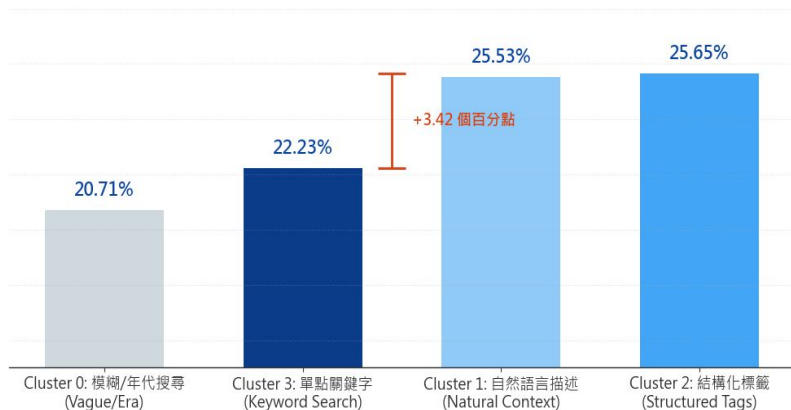
Cluster 3: Keyword Search

「周杰倫」、「搖滾」、「安靜」

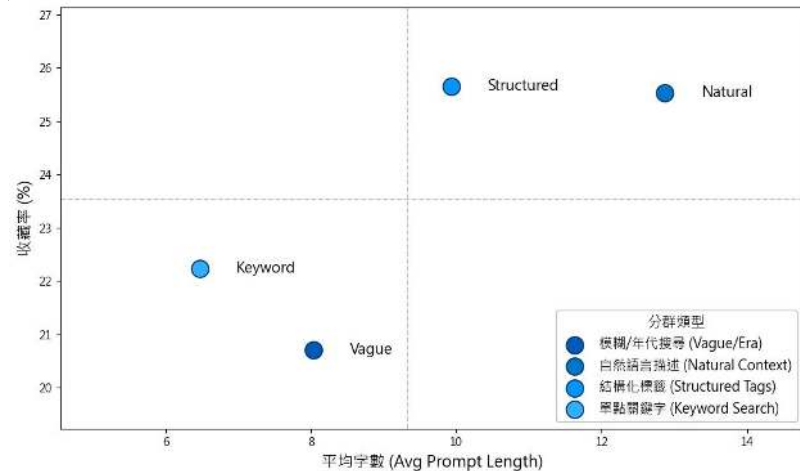
N = 18,812

Natural Context and Structured Tags provide clearer context, enabling AI to deliver more relevant recommendations

Structured tags have the highest CR



Keywords and word count affect bookmarking



Most users treat AI playlists as a 'smarter search bar,' obscuring the feature's true value

User Thought

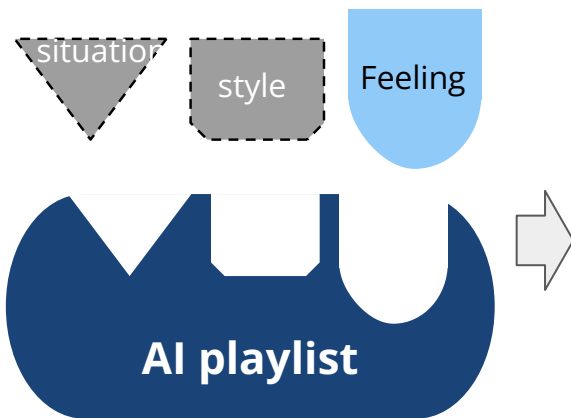
Shorter is better
Single keyword
Fast result return

VS.

What AI Needs

The more complete the context and preferences, the better the recommendation.

User behavior differs from AI requirements



Expectation gap
Low efficiency
Decreased perceived value

The core issue isn't AI performance, but user behavior failing to align with Generative AI best practices.

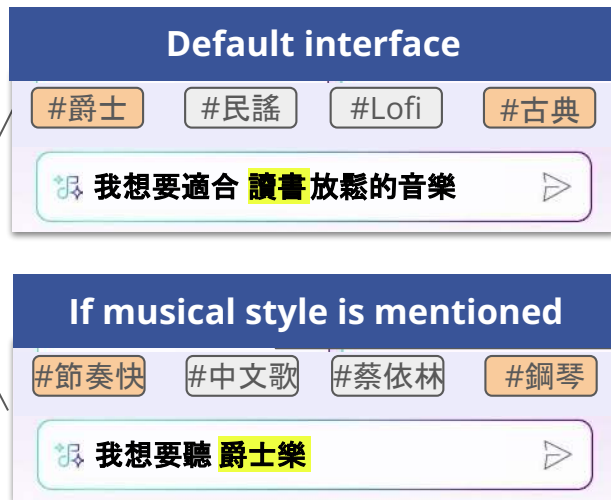
04 Solution I: Import Guided Prompt Interface



Guided Prompting: Use 'Genre + Personalized Hashtags' and dynamic filtering to improve input accuracy.

'Genre + Personalized Hashtags' and dynamic filtering

- Focusing hashtags on genres to address user pain points, displaying only sub-tags
- Apply dynamic filtering based on initial prompts to show only highly relevant options



Recommendation logic

Default Interface:

4*Music Genres

+ 2*Personal Recommendations

If the music genre is mentioned:

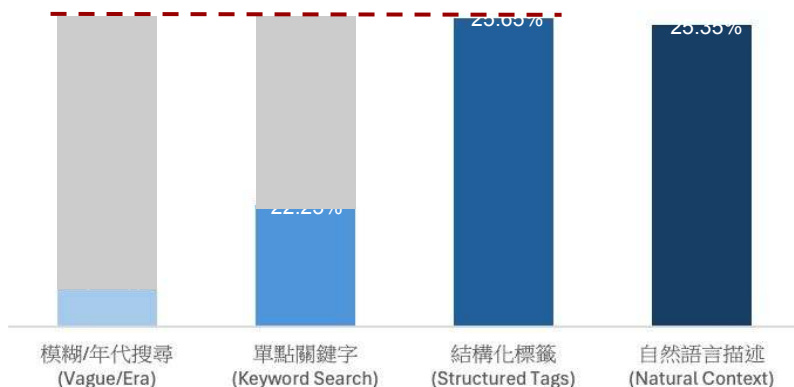
6*Personal recommendations

Interaction: Auto-display hashtags based on input after a 600ms debounce.

If vague prompts matched the save and retention rates of clear inputs, it would yield 972 extra saves (+5.7%) and 285 extra returns (+1.4%), boosting accuracy and satisfaction.

By using a guided Prompt interface, structured usage behavior is encouraged, increasing CR

CR of each prompt behavior pattern



Structured tags yield the highest save rates, while vague and single-keyword inputs show the largest performance gap

Through guided prompting, we aim to elevate the save rates of 'vague/era search' and 'single-keyword' inputs to the level of 'structured tags'

It is expected to increase the total number of collections to

11,674 collection

The total number of songs in the collection is expected to increase to

1.89 million songs

04 Solution II: KKBOX AI New Positioning



Transforming the feature to fully leverage AI potential and pivot user habits towards AI-native behaviors

Strategic goals and execution entry points

- Introduce new AI scenarios to gradually pivot user habits, evolving AI from a search tool into an interactive companion
- Leverage existing interactive features—AI Playlists and 'Listen With'—as entry points to strengthen the AI's persona

Transforming AI playlist positioning

AI Music Consultant

Focusing on 'immediacy' and 'personalization' to proactively offer micro-recommendations based on individual traits

Proactive
notifications

Close to life

Emotional

'AI Tool' to a 'Music Recommendation Assistant'

Enhancing the favorability and value of AI

AI Music Radio Assistant

Leverage KKBOX's existing social features as entry points to enhance the affinity and perceived value of AI Playlists.

Host

Followers

AI serves as more than just a functional tool; it offers companionship like a friend.

The AI Music Consultant precisely identifies the right timing for music needs, delivering recommendations centered on the individual user

Difference	Focusing on 'immediacy' and 'personalization' to proactively offer micro-recommendations based on individual traits, fostering deeper emotional resonance
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- To avoid notification fatigue, the system prioritizes micro-recommendations during peak user activity, reducing cognitive load and mitigating the risk of negative feedback from poor suggestions.
- Enhances the 'consultant' persona; this requires strengthening the model's emotional semantic understanding to respond more precisely to user needs.

AI Advisor Core Functions

Recommend Content	Resource	Example
Context-triggered (immediacy-based)	Link to external resources	Typhoon, rain, and other times
Personalized recommendation (personal feeling)	Detect user activity on the app	Search for songs with vastly different musical styles or Unusual opening late at night

AI Music Consultant Usage Process

Triggered Notifications

Real-time context awareness



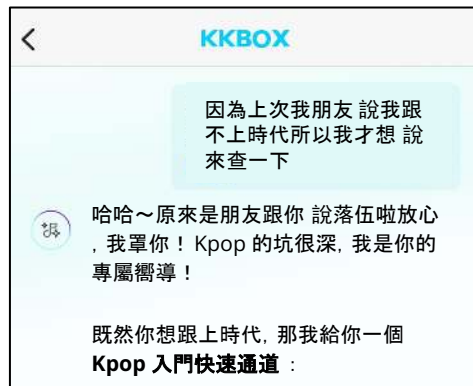
Existing AI playlist Feature Interface

Provide reasoned recommendations to build trust



Continuous Dialogue

Two-way learning



AI Station Assistant: Enhances AI's affinity and value through personalized replies in familiar social settings

Focusing on the KKBOX "Listen With" community

Enhance AI interactivity
by leveraging a
social-driven approach

Challenges in establishing an
AI-centric social platform

Optimize the existing and familiar 'Listen
With' social community

KKBOX's current function

太和路快速

Bye

2025/9/28 下午 4:51

Ru Ru

非常喜歡的樂團歌曲

2025/10/13 下午 3:17

jennie

Like Jennie



- Low willingness
- Impersonal chatroom

The social potential of AI

Yusyuan Ye

YY 好酷喔

ChatGPT

收到~聊天情境更新:「第一次見面很緊張」x2, 氣氛感覺更明顯了 😊

來一首能把緊張感化解成舒服氛圍的歌:

🎵《突然好想你》—五月天 (輕柔版氛圍)

旋律溫和、有熟悉感, 能讓大家在第一次見面時比較放鬆、自然, 不會太吵也不會太安靜。

想轉換成更活潑、搞笑、或 chill 氛圍也可以, 繼續聊天我就會跟著調整歌單 🎧👉

- Utilize real-time AI responses to sustain chat momentum

**Integrate AI into
familiar interactive
scenarios to enhance
affinity and perceived
value**

Dual-sided AI Assistance: Lowering barriers for Hosts and fostering companionship for Followers

Feature Design: Integrating AI into "Listen With"

Host

Lower the barriers to entry and enhance motivation

- The AI suggests 2-3 'next-up' tracks by analyzing attendee requests, chat sentiment, etc.
- Provides real-time insights for the current track, including song background, lyrical highlights, etc.

Follower

Fostering companionship to drive retention and user activity

- Respond to user chats in real-time, such as answering questions or welcoming newcomers.
- Set conditional triggers for AI replies to prevent over-engagement



'Depth-Oriented' and 'Breadth-Oriented' approaches based on user cluster differences

	Breadth-oriented	Depth-oriented
	Core User	Trial User
	Scattered User	Potential User
AI music consultant	Using a companion role to trigger exploratory resonance	A specialist role offering expert taste in specific musical genres
AI Music Radio Assistant	A 'Mood & Scenario' theme that spans across multiple genres "A pre-sleep chat" or "A focused plan for rainy days"	Showing AI's depth of knowledge through 'genre-themed' topics "K-POP Night" or "Rap Mania Week"

05 Summary



From experience optimization to strategic AI repositioning

Pain points Solutions		AI - Prompt understanding gap	Using AI as a general search bar	Project Objective: Improve Satisfaction (CR, RR)
Guided Prompt		Assist users in crafting detailed prompts	Move beyond single keywords	Enhance to reach 11,674 collections and reduce one-time churn
KKBOX AI new positioning	music consultant	Encourage users interact through natural language	Strengthen the "Consultant" persona	Reposition AI to shift user habits, improving response accuracy and DAU (Daily Active Users)
	Music Radio Assistant	Leverage conversational interfaces to encourage users to interact using natural language	Create emotional bonds and community belonging	

Thank you

KKBOX

Appendix

01 | User Segmentation EDA

02 | Prompt Analysis

03 | Data Cleaning & EDA

Appendix 01

User Segementation EDA



Overview of User Segmentation Behavior

	Index	Core (6.4%)	Scattered (10.7%)	Potential (28.3%)	Trial (54.6%)
Group Overview	Avg. Single-use Intensity	2.85	1.25	2.86	1.01
	Avg. Collection Rate (CR)	41.7%	31.3%	33.0%	16.5%
	Effective Users (%)	63.3%	47.0%	33.3%	16.6%
	Contribution of CR	10.9%	13.7%	38.2%	37%
First-time user experience	First Stay Time	3.30 mins	0.52 mins	2.67 mins	0.02 mins
	First Use Prompt Count	3.06	1.26	2.88	1.01
	Avg Conversion Rate	39.9%	32.5%	27.6%	16.0%
	Model Response Time (sec)	10,421	10,244	10,516	9,190
	Avg. Song Count	16.75	17.38	16.88	17.52

Top 10 songs played by user segment

Core (1,631 people)

Classical, Sleep, Anime Theme Songs

1. 晚安曲(布拉姆斯的搖籃曲) - 21,912 次
2. HOT - 13,955 次
3. 龍貓-選自《龍貓》- 12,892 次
4. 聽見睡眠氛圍(下雨聲) - 12,251 次
5. Golden - 10,971 次
6. 看著我的眼睛說 - 10,180 次
7. Soda Pop - 8,649 次
8. Jumping Machine (跳樓機) - 8,547 次
9. 白噪聲, Pt. 1 - 7,750 次
10. Your Idol - 6,639 次

Potential (6,905 people)

Popular songs

1. Say Less - 45,333 次
2. Golden - 39,792 次
3. Express Mode - 39,139 次
4. 看著我的眼睛說 - 35,397 次
5. Haircut - 32,846 次
6. Air - 32,836 次
7. Finale - 32,199 次
8. 我們的花語 (Stuck With You) - 32,025 次
9. I Know - 31,832 次
10. Delight - 31,412 次

Scattered (2,383)

Classical, Animation, Children's Songs

1. 布拉姆斯 — 搖籃曲 作品49 第4號 - 26,949 次
2. Golden - 17,264 次
3. 看著我的眼睛說 - 17,182 次
4. 瑟魯之歌 - 15,836 次
5. Jumping Machine (跳樓機) - 11,974 次
6. 沒去過的地方 (Never Been) - 11,853 次
7. Sorry I'm Here For Someone Else - 11,764 次
8. Soda Pop - 11,458 次
9. Express Mode - 11,205 次
10. Baby Shark - 8,113 次

Trail (12,531 people)

Popular songs

1. 看著我的眼睛說 - 82,635 次
2. Golden - 74,931 次
3. Jumping Machine (跳樓機) - 69,897 次
4. Express Mode - 53,300 次
5. Soda Pop - 45,811 次
6. Air - 43,351 次
7. Haircut - 42,715 次
8. 座位 - 41,069 次
9. 沒去過的地方 (Never Been) - 40,813 次
10. I Know - 40,327 次

Detailed information on user segmentation depth and breadth

Segment	User Count	Avg. Play Frequency	Median Play Count	Total Plays	Avg. Different Songs Count	Median Different Songs Count	Avg. Repeat Rate	Avg Single-Track Repeat Count
Scatteres	2383	3399.73	2150.0	8101545	1117.3	732.0	0.43	4.02
Trail	12531	2582.83	1535.0	32365440	787.0	510.0	0.43	4.42
Core	1631	3041.5	1840.0	4960688	937.44	623.0	0.43	3.81
Potential	6905	2467.88	1515.0	17040691	770.24	512.0	0.42	4.32

Appendix 02

Prompt Analysis



Semantic characteristics of high-collection and high-retention prompts: Clear intent and structured semantics.

Prompt_Tag	Prompt CR	Prompt RR	Semantic attributes
純音樂	31.90%	32.91%	明確屬性、AI 易辨識
人聲	29.70%	34.25%	可直接對應資料庫
節奏	26.60%	28.93%	可量化條件
藝人	26.30%	26.40%	強目的導向
排除 (如「不要悲傷的歌」)	26.30%	28.42%	條件式語意明確
語言	26.20%	30.05%	可直接對應分類
地區	25.60%	30.06%	有清楚篩選邏輯
曲風	25.40%	28.49%	主流屬性標籤
年代	24.40%	28.51%	有明確時間範疇

Appendix 03

Data cleaning & EDA



Data Fields Overview

Column Name	ID	Date	Prompt	Response Tags	Prompt Tags	Score	Add to CPL	User	Response Time	Song Count
Description	唯一識別碼	日期與時間	輸入的指令	系統回應的分類標籤	內容的分類標籤	評分, 預設為空值	是否新增至個人歌單	編碼過的用戶識別碼	歌單內容時間(毫秒)	歌單中的歌曲數量
Ex.	646377	2025-06-18 00:00:40	歌德重金屬風	AI Search	曲風	1	TRUE	0llctfp8vxaRRy15Aq	9036	10

Data Cleaning

Origin

n = 57990

Pre-processing

Data standerdize and type change

Missing Value Check

Score :95.75%、Prompt_Tags:0.01%

Missing Value Handling

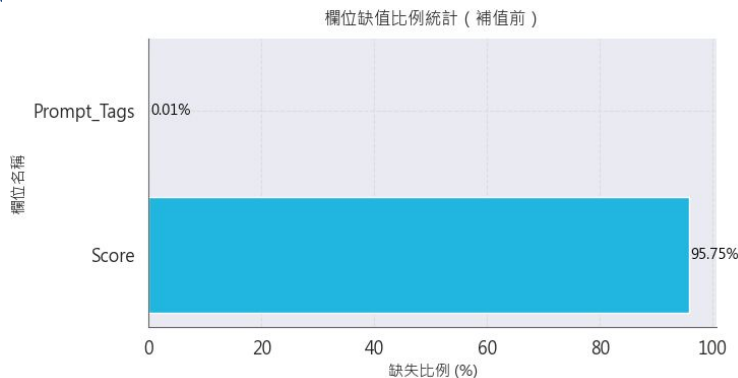
Score padded with zeros、Prompt_Tags retain

Outlier Check

Response_Time range too large (0 milliseconds / >20000 milliseconds)
→ Reserved

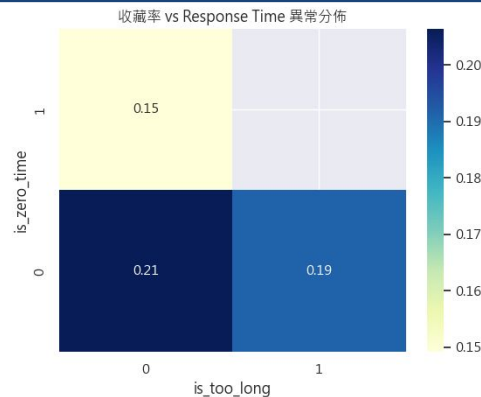
Data Cleaning

Missing Value Check



- Only 3 Prompt_Tags are missing, and this has no impact on other columns.
- Missing Score values are filled with 0.a

Outlier Check



- Response_Time does not affect the collection rate → Retain